energy/low-vibration-energy or low-winding-energy/high-vibration-energy—are physically indistinguishable. And, unlike the financial analogy for which considerations beyond total wealth can distinguish between the two types of stock holdings, there is absolutely no physical distinction between the two string scenarios.

Actually, we shall see that to make the analogy with string theory tighter, we should consider what would happen if you did not divide your money equally between the two companies in your initial investment, but bought, say, 1,000 shares of the fitness company and 3,000 shares of the heart-valve company. Now the total value of your holdings does depend on which company closes high and which closes low. For instance, if the stocks close at $10 (fitness) and 10 cents (heart-valve), your initial investment of $4,000 will now be worth $10,300. If the reverse happens—the stocks close at 10 cents (fitness) and $10 (heart-valve)—your holdings will be worth $30,100 — significantly more.

Nevertheless, the inverse relationship between the closing stock prices does ensure the following. If a friend of yours invests exactly "opposite" to you—3,000 shares of the fitness company and 1,000 shares of the heart-valve company—then the value of her holdings will be $10,300 if stocks close valves-high/fitness-low (the same as your holdings in the fitness-high/valves-low closing) and $30,100 if they close with fitness-high/valves-low (again, the same as your holdings in the reciprocal situation). That is, from the point of view of total stock value, interchanging which stock closes high and which closes low is exactly compensated by interchanging the number of shares you own of each company.

Hold this last observation in mind as we now return to string theory and think about the possible string energies in a specific example. Imagine that the radius of the circular Garden-hose dimension is, say, ten times the Planck length. We will write this as $R = 10$. A string can wrap around this circular dimension one time, two times, three times, and so forth. The number of times a string wraps around the circular dimension is called its winding number. The energy from winding, being determined by the length of wound string, is proportional to the product of the radius and the winding number. Additionally, for any amount of winding, the string can undergo